

Slaughter & Butchery Module

System Overview

Uganda Livestock Information Tracking System (U-LITS)

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Abstract

This document provides a comprehensive overview of the Slaughter and Butchery module within the Uganda Livestock Information Tracking System (U-LITS). The module tracks the complete livestock processing chain from farm to consumer, ensuring end-to-end traceability, quality assurance, and regulatory compliance for meat production.

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1 Introduction

The Slaughter and Butchery module tracks the complete livestock processing chain from farm to consumer. It provides end-to-end traceability, quality assurance, and regulatory compliance for meat production.

1.1 Process Overview

The complete workflow follows this sequence:

Movement Permit → Slaughter House → Carcass Processing → Quarters → Meat Cuts → Final Products → Sale

2 Movement Permit

2.1 Purpose

Movement Permits authorize and document the transport of livestock from farms to slaughter houses or between farms. This ensures proper documentation and tracking during transit.

2.2 Process

The movement permit process follows these steps:

1. Farmer or trader applies through mobile app or web portal
2. Selects animals by E-ID (Electronic ID) or V-ID (Visual ID)
3. Specifies destination (slaughter house or farm)
4. District Veterinary Officer reviews and approves or rejects the application
5. System generates permit with unique number and QR code
6. Animals are transported with accompanying permit document

2.3 Key Information Tracked

The system tracks the following information for each permit:

- Origin and destination locations
- Animal identification (E-ID, V-ID, breed, sex, age)
- Transport details (vehicle, route, checkpoints)
- Validity period
- Approval status

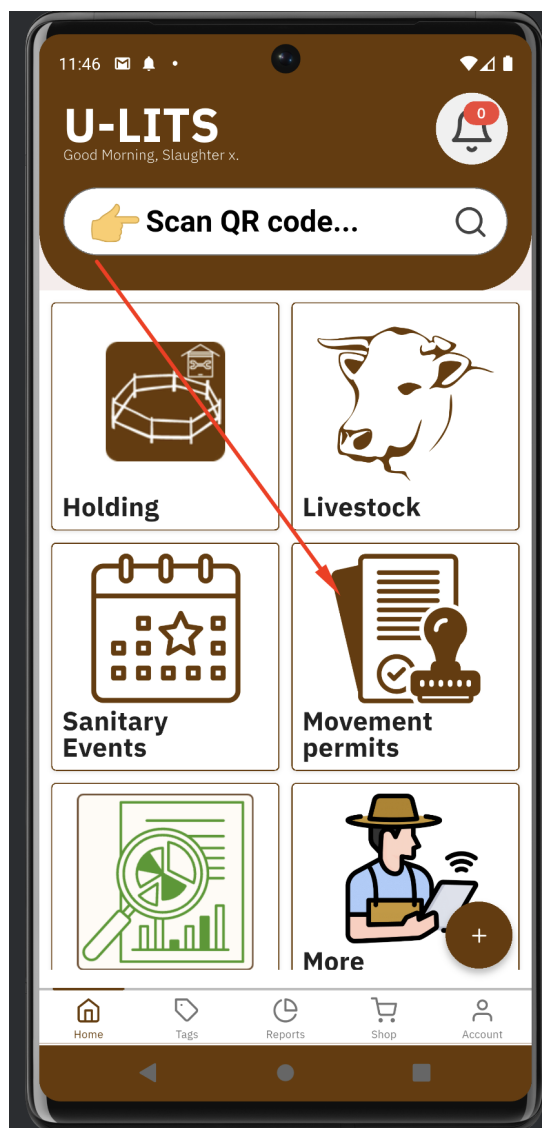


Figure 1: Accessing Movement Permit Module from Home Page

2.4 Requirements

- Animals must have proper identification tags
- Health inspections required before movement
- Permit approval mandatory before transport
- Animal list cannot be modified during transit
- Subject to checkpoint inspections

3 Slaughter House Operations

3.1 Animal Reception

Upon arrival at the slaughter house, staff verify:

- Movement permit is valid and approved

11:53

← Movement permit ✓

STEP 2/3

- **Livestock's destination info. (TO)**

Select movement

To farm ✓ To slaughter Other

Pick slaughter house

Kalerwe

- **Transportation info.**

Transporter's name

Vehicle's reg' no.

Transporter's NIN

Transporter's Phone

NEXT →

Figure 2: Movement Permit Application Form - Selecting Destination Slaughter House

- Animals match permit details
- Animals are fit for slaughter

3.2 Antemortem Inspection

A veterinary officer examines live animals for:

- Overall health condition
- Disease signs or injuries
- Fitness for slaughter
- Unfit animals are rejected

3.3 Slaughter Process

The slaughter process includes:

- Only approved animals proceed
- Barcode generated for carcass tracking
- Humane slaughter practices followed
- Slaughter record created in system

3.4 Postmortem Inspection

Following slaughter, the veterinary officer examines the carcass for:

- Disease indicators in organs
- Meat quality assessment
- Contamination or abnormalities
- Meat grading (Prime, Choice, Standard, etc.)

3.5 Recording

The system captures comprehensive information:

- Animal identification (E-ID, V-ID, LHC)
- Pre-slaughter inspection findings
- Post-slaughter inspection results
- Carcass weight and grade
- Slaughter date and location
- Barcode and QR code for traceability

4 Butchery Process ---

The butchery process transforms whole carcasses into sellable meat products through three distinct stages.

4.1 Stage 1: Quartering (Primary Breakdown)

The carcass is divided into four main sections:

- **Fore-Right (FR)** – Front right portion
- **Fore-Left (FL)** – Front left portion
- **Hind-Right (HR)** – Rear right portion
- **Hind-Left (HL)** – Rear left portion

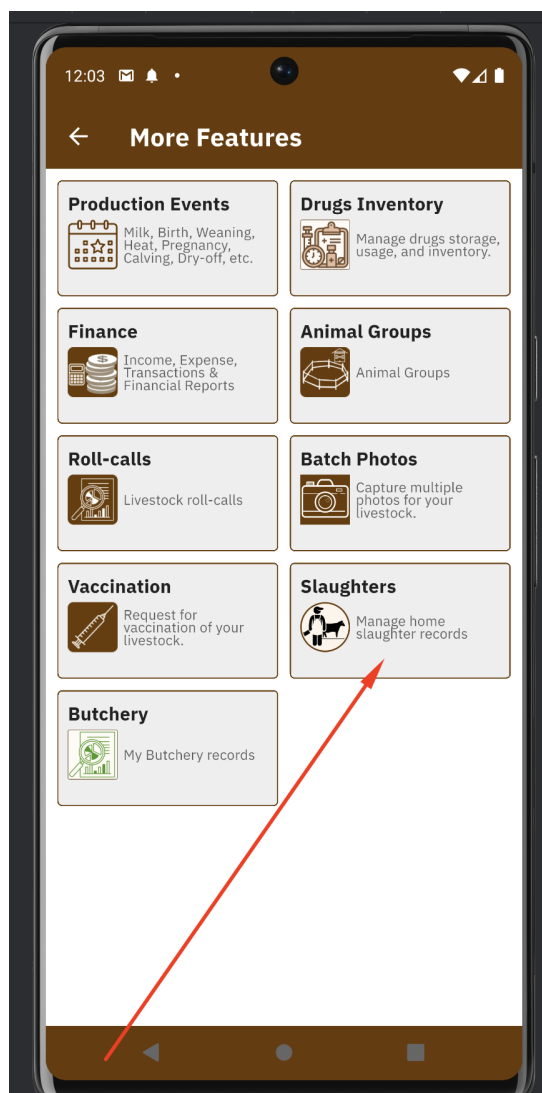


Figure 3: Slaughter House Officer Accessing Slaughter Records Module

4.1.1 Quarter Processing

Each quarter is:

- Weighed individually
- Recorded as a Slaughter Distribution Record
- Tracked for remaining available weight
- Assigned for further processing

The sum of quarter weights should approximate the original carcass weight (within 5% tolerance).

4.2 Stage 2: Cuts (Secondary Breakdown)

Each quarter is further divided into specific cuts classified as either Primal Cuts or Offal Cuts.

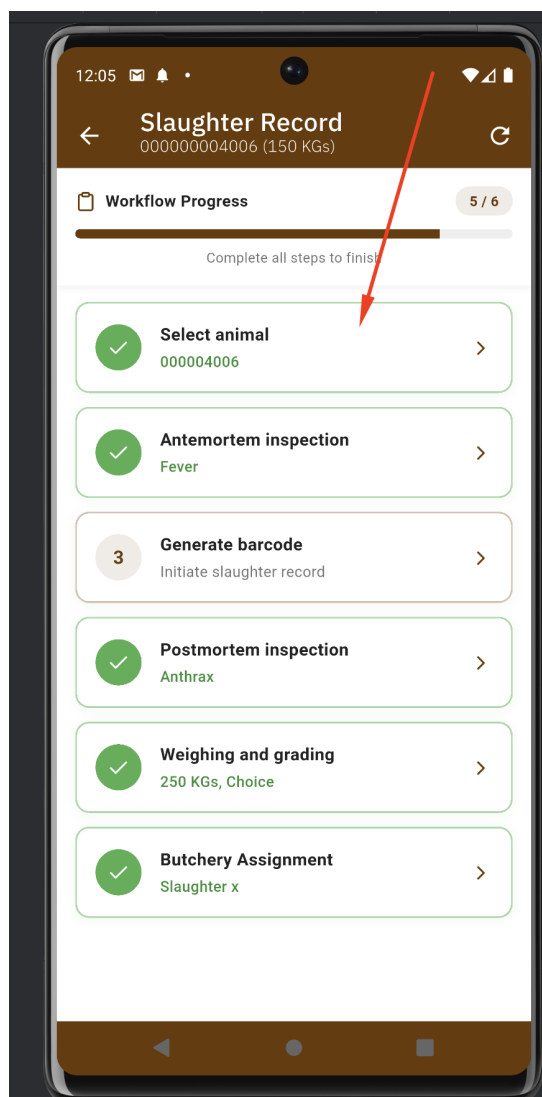


Figure 4: Creating New Slaughter Record from Movement Permit

4.2.1 Primal Cuts (29 types)

Main meat portions including:

- Sirloin/Striploin, T-Bone, Fillet, Rib Eye
- Rump, Brisket, Chuck, Topside
- Family Steak, Minced Meat, Bones
- And 18 additional cut types

4.2.2 Offal Cuts (4 types)

Organ meats including:

- Liver
- Heart
- Kidneys
- Tongue

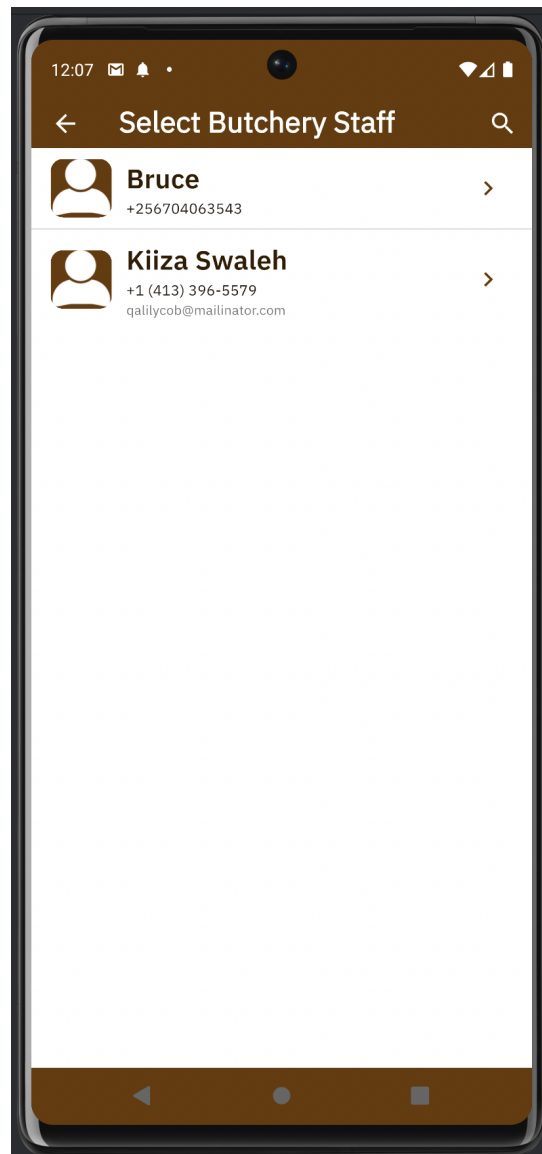


Figure 5: Assigning Carcass to Butchery After Slaughter Process

4.2.3 Cut Recording

Each cut is recorded with:

- Source quarter (FL, FR, HL, HR)
- Cut classification (Prime or Offal)
- Specific cut name
- Weight in kilograms
- Barcode for tracking

The system automatically reduces the quarter's available weight when cuts are recorded.

4.3 Stage 3: Final Packaging (Butcher Records)

Cuts are packaged into consumer-ready portions:

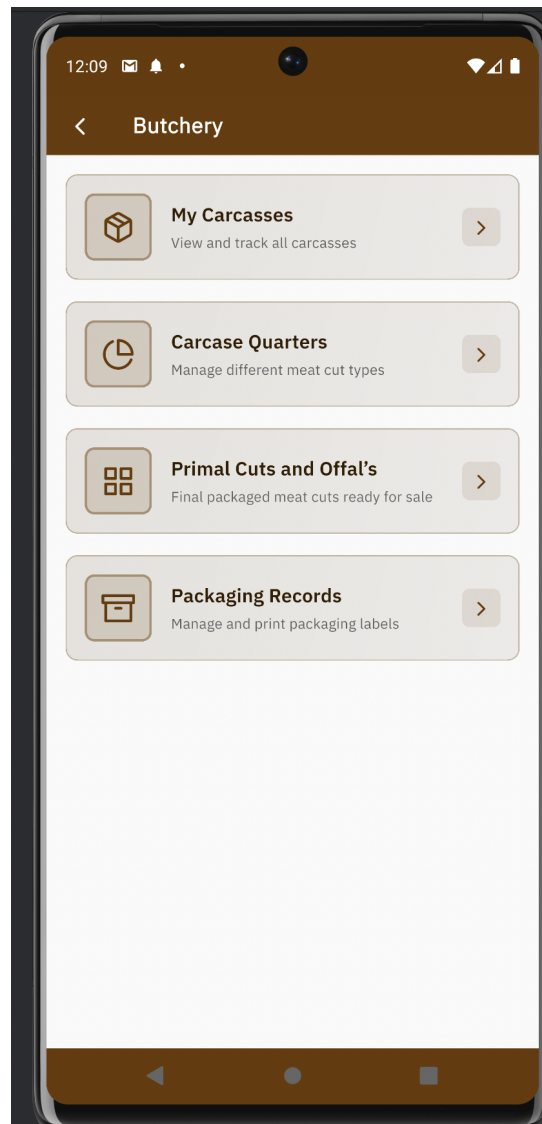


Figure 6: Butchery Officer Accessing Butchery Module

- Individual package weight recorded
- Price assigned
- Barcode and QR code generated
- Status marked (Available or Sold)
- Full traceability maintained

4.3.1 Sales Recording

When sold, the system records:

- Status updated to “Sold”
- Buyer information recorded
- Sale price and date captured
- Inventory automatically adjusted

The image shows a mobile application interface for recording carcass quarters. The screen has a dark blue header with a back arrow, the title 'Record Carcase Quarters', and a subtitle '000004157 (100 KGs)'. Below the header, there are three input sections: 'Select Carcass *' with a text field containing '000004157 - KGs (Grade: Prime)', 'Cut section' with four radio button options ('Fore-1/4 - Right', 'Fore-1/4 - Left', 'Hind-1/4 - Right' (selected), and 'Hind-1/4 - Left'), and 'Quantity (Kgs) *' with a text field containing '50'. At the bottom of the form is a large blue 'Submit' button. The phone's status bar at the top shows the time 12:11 and various icons.

Figure 7: Creating Quarters from Assigned Carcass

5 Data Relationships

5.1 Weight Flow

The weight tracking flows through the following hierarchy:

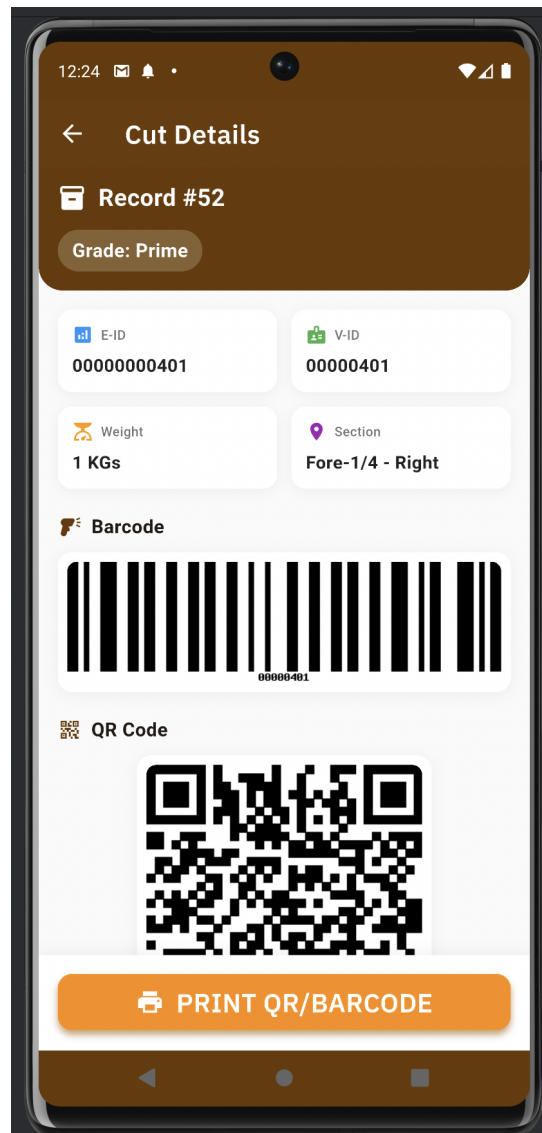


Figure 8: Quarter Barcode and QR Code Label for Traceability

```

Slaughter Record (Carcass)
|-- Available Weight: Total carcass weight
|
v
Slaughter Distribution Records (Quarters)
|-- 4 quarters created
|-- Carcass available weight reduced to 0
|
v
Slaughter Distribution Records (Cuts)
|-- Multiple cuts per quarter
|-- Quarter available weight reduced accordingly
|
v
Butcher Records (Final Products)
|-- Packaged portions from cuts
|-- Ready for sale with full traceability

```

12:24

< Create Butcher Record
00000401 (1 KGs)

Select Meat Cut *
00000401 - 1 KGs (Grade: Prime)

Weight (Kgs) *

Price (UGX) - Optional

Cut Type *
☒ Prime Cut
☐ Offal Cut

Prime Cut Type *

Sale Status *
☒ Not Sold ☐ Sold

Notes (Optional)

Submit

Figure 9: Creating Prime Cuts and Offal Cuts from Quarters

5.2 Practical Example

5.2.1 Initial Slaughter

- 250 kg carcass recorded

5.2.2 After Quartering

- Fore-Right: 62 kg
- Fore-Left: 63 kg
- Hind-Right: 65 kg
- Hind-Left: 60 kg
- Total: 250 kg (matches carcass weight)

5.2.3 Processing Fore-Right Quarter (62 kg)

- Sirloin: 15 kg

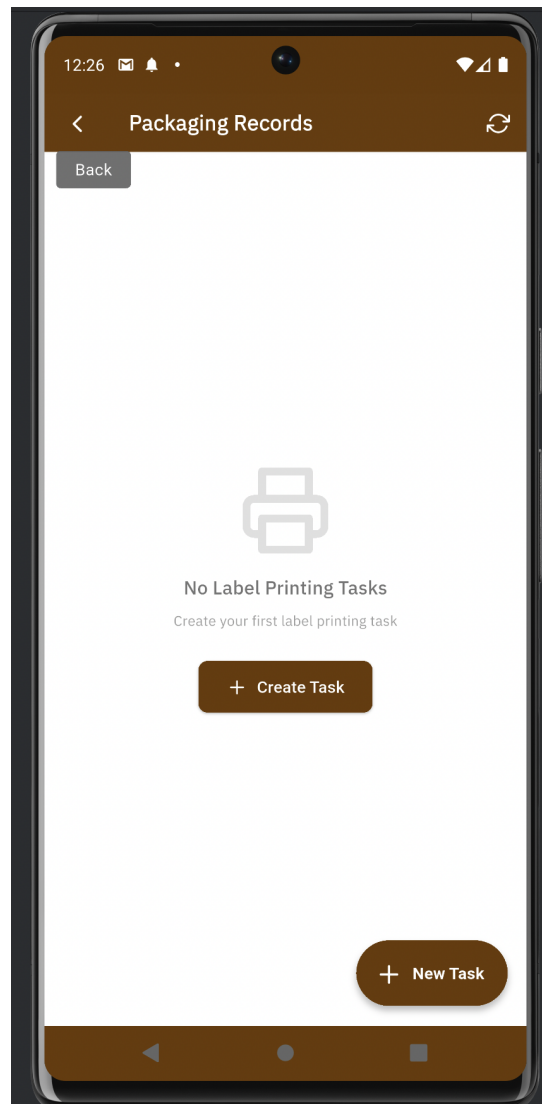


Figure 10: Creating Final Packaging with Labels and Barcodes

- T-Bone: 8 kg
- Chuck: 20 kg
- Bones: 10 kg
- Liver: 2 kg
- Heart: 2 kg
- Waste/Trim: 5 kg
- Total: 62 kg (quarter fully processed)

5.2.4 Final Packaging

- 15 packages of Sirloin (1 kg each)
- 8 packages of T-Bone (1 kg each)
- 20 packages of Chuck (1 kg each)

- Each with unique barcode/QR code

6 User Roles ---

6.1 Farmer/Trader

- Applies for movement permits
- Selects animals for transport
- Tracks permit status

6.2 District Veterinary Officer

- Reviews permit applications
- Approves or rejects permits
- Monitors district animal movements

6.3 Checkpoint Inspector

- Verifies permits during transit
- Inspects animals at checkpoints
- Records checkpoint passage

6.4 Slaughter House Officer

- Receives animals with permits
- Conducts inspections
- Creates slaughter records
- Assigns meat grades

6.5 Butcher

- Records quarters and cuts
- Manages inventory
- Creates final packaged products
- Records sales transactions

6.6 System Administrator

- Registers facilities
- Manages user accounts
- Monitors compliance
- Generates reports

7 System Features

7.1 Mobile Application

The mobile application provides:

- Movement permit management
- Slaughter record creation
- Inspection form completion
- Barcode generation and scanning
- Quarter and cut recording
- Sales transaction recording
- Full traceability access

7.2 Web Portal

The web portal offers:

- Comprehensive dashboard with statistics
- Record management (slaughters, distributions, butcher records)
- Advanced filtering and search capabilities
- Inventory tracking
- Sales reporting
- Label generation
- Data export capabilities

8 Traceability

Every meat product maintains complete traceability through the supply chain:

```
Consumer Product (Butcher Record)
|
v
Meat Cut (Slaughter Distribution Record)
|
v
Quarter (Slaughter Distribution Record)
|
v
Carcass (Slaughter Record)
|
v
Live Animal (Movement Permit)
|
v
Farm of Origin
```

8.1 QR Code Information

Scanning a QR code on any package reveals:

- Farm and farmer details
- Movement permit information
- Slaughter house and date
- Veterinary inspection results
- Meat grade and weight
- Processing history
- Complete audit trail

9 Quality Assurance

Quality control occurs at multiple stages throughout the process:

1. **Farm Level** – Animal health records and vaccination status
2. **Transport** – Movement permit verification and checkpoint inspections
3. **Reception** – Permit validation and initial assessment
4. **Antemortem** – Live animal veterinary inspection
5. **Postmortem** – Carcass and organ examination
6. **Grading** – Meat quality classification
7. **Processing** – Weight verification and hygiene standards
8. **Packaging** – Proper labeling and traceability codes
9. **Sales** – Transaction recording and buyer documentation

10 Business Rules

10.1 Movement Permits

- Minimum one animal required
- All animals must have identification
- Destination must be specified
- Immutable after approval
- Expires after validity period

10.2 Slaughter Records

- Created only from approved permits
- Antemortem inspection mandatory
- Postmortem required before grading
- Cannot complete until all weight distributed

10.3 Quarters

- Must reference valid carcass
- Total weights should match carcass ($\pm 5\%$)
- Four distinct quarters per carcass
- Cannot exceed available carcass weight

10.4 Cuts

- Must reference valid quarter
- Weight cannot exceed quarter availability
- Requires cut type specification
- Automatically reduces quarter weight

10.5 Butcher Records

- Created from valid cuts
- Cannot exceed source cut weight
- Prime or Offal classification required
- Sold status is permanent

11 System Benefits

11.1 For Farmers

- Official transport documentation
- Access to formal markets
- Fair pricing based on grades

11.2 For Slaughter Houses

- Digital record keeping
- Regulatory compliance
- Quality control systems
- Efficient operations

11.3 For Butchers

- Inventory management
- Sales tracking
- Professional labeling
- Accurate traceability

11.4 For Consumers

- Origin transparency
- Quality verification
- Food safety assurance
- Veterinary certification access

11.5 For Government

- Disease surveillance
- Food safety monitoring
- Market regulation
- Policy-making data

12 Conclusion

The Slaughter and Butchery module provides a comprehensive digital solution for meat production management. By connecting all stakeholders in the supply chain and maintaining complete traceability from farm to consumer, the system ensures food safety, regulatory compliance, and market transparency.

This integrated approach strengthens Uganda's livestock sector by providing transparent, accountable, and efficient meat production processes that benefit all participants in the value chain.